

Seungheon Lee

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Work Experience

Applied Intuition

Seoul, S.Korea

Application Engineer

May 2026 - Present

- Collaborated with autonomy developers to design scalable software solutions, adapting products to accelerate customers' Physical AI and autonomous mobility initiatives.

Hyundai Motor Company

Gyeonggi, S.Korea

Research Engineer (Technical PM)

Jan. 2025 - May 2026

- Led project management for a large-scale (600+ personnel) Robotaxi production project.
- Minimized risks to critical milestones (e.g., certification, firmware deployment) by proactively negotiating test vehicle production capacity and implementing a priority-based resource allocation strategy.
- Collaborated with legal and security departments to review the consistency of project deliverables, proactively identifying and addressing potential regulatory and security risks.

Samsung Research

Seoul, S.Korea

Software Engineer Intern

Jul. 2024 - Aug. 2024

- Developed a monitoring system for LLM API servers, addressing previously unimplemented functionalities for an upcoming service launch.
- Built Kubernetes clusters and set up a Dev server to support scalable infrastructure and system development.
- Collaborated with administrators to identify and implement key features, including detailed API analytics and real-time alert systems.

Robotics Lab, University of Seoul

Seoul, S.Korea

Undergraduate Researcher

Mar. 2022 - May. 2024

- Grand Prize**, Autonomous Robot Race Competition: Led a software architecture overhaul; this refactoring from a monolithic script to a modular framework established the robust foundation required to implement advanced algorithms, including Model Predictive Control (MPC).
- First Place**, RoboCup Autonomous Robot Manipulation Challenge: Engineered a high-fidelity training dataset for the winning model by implementing a precise polygon labeling method and a novel data pipeline benchmarked from international competitors.
- Presented 3 research papers (2 posters, 1 oral) at academic conferences based on competition findings.

Education

University of Seoul

Seoul, S.Korea

B.S. in Mechanical Information Engineering, Top Graduate

Mar. 2019 - Feb. 2025

- GPA: 4.15 / 4.5 (Major GPA: 4.35 / 4.5)

Skills

Languages	Python, C/C++
AI & MLOps	PyTorch, Weights & Biases (W&B), MLflow, Grafana
Infrastructure & Tools	Linux, Kubernetes, Docker, MATLAB
Hardware & Sensors	LiDAR (2D, 3D), GPS, RGB-D Camera

Projects

Computer Vision AI Model Optimization for NPU-based Inference

Sep. 2024 - Dec. 2024

- Collaborated in a 2-member team for the 1st AI Semiconductor Competition, winning the **Grand Prize** (Minister of Science and ICT Award).
- Achieved up to 533% improvement in NPU utilization compared to the baseline, while simultaneously increasing quantization accuracy.
- Established an MLOps pipeline using WandB and MLflow for systematic experiment tracking and model management.

AI-powered Smart Bartending System

Mar. 2024 - Oct. 2024

- Architected and implemented the core AI engine to translate ambiguous user inputs (e.g., mood, taste) into quantitative cocktail recipes.
- Leveraged prompt engineering and few-shot learning to robustly handle qualitative data uncertainty, bypassing the need for extensive model fine-tuning.

Awards & Scholarships

AWARDS

2024	Grand Prize (Minister of Science and ICT Award) , 1st AI Semiconductor Competition	Seoul, S.Korea
2023	First Place , RoboCup Autonomous Robot Manipulation Challenge, hosted by MathWorks	Bordeaux, France
2023	Grand Prize , Autonomous Robot Race Competition	Cheongju, S.Korea
2023	Excellence Award , Engineering Research Internship Outcome Presentation	Seoul, S.Korea

SCHOLARSHIPS

2023 - 2025	OK Scholarship , OK Scholarship Foundation
2023 - 2024	Blue Lighthouse KB Dream Wave 2030 Promising Field Scholarship , Korea Scholarship Foundation